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| **Fiscal Year:** | FY 2019 | **Task Last Updated:** | FY 05/02/2019 |
| **PI Name:** | Williams, Jason Ph.D. | | |
| **Project Title:** | Fundamental Interactions for Atom Interferometry with Ultracold Quantum Gases in a Microgravity Environment | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | FUNDAMENTAL PHYSICS--Fundamental physics | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **PI Organization Type:** | NASA CENTER | **Phone:** | 303-725-1580 |
| **Organization Name:** | NASA Jet Propulsion Laboratory | | |
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| **City:** | Pasadena | **State:** | CA |
| **Zip Code:** | 91109-8001 | **Congressional District:** | 27 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2013 Fundamental Physics NNH13ZTT002N (Cold Atom Laboratory--CAL) |
| **Start Date:** | 04/01/2014 | **End Date:** | 05/03/2021 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA JPL |
| **Contact Monitor:** | Callas, John | **Contact Phone:** | |
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| **Flight Program:** | ISS | | |
| **Flight Assignment:** | ISS<br>NOTE: End date changed to 5/3/2021 per PI information (Ed., 5/6/19) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | D'Incao, Jose Ph.D. ( University of Colorado )<br>Elliott, Ethan Ph.D. ( Jet Propulsion Lab ) | | |
| **Grant/Contract No.:** | Internal Project | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | Precision atom interferometers (AI) in space promise exciting technical capabilities with diverse applications of interest to NASA. These quantum sensors are particularly relevant for fundamental physics research, with proposals including unprecedented tests of the validity of the weak equivalence principle, precision measurements of the fine structure and gravitational constants, and detection of gravity waves and dark matter/dark energy. Our studies will utilize the capabilities of NASA's multi-user Cold Atom Laboratory (CAL), in the microgravity environment onboard the International Space Station (ISS), to study mitigation schemes for the leading-order systematics expected to limit future high-precision measurements of fundamental physics with AIs in microgravity. The flight experiments, supported by theoretical investigations and ground studies at our facilities at Jet Propulsion Laboratory (JPL), will concentrate on the physics of pairwise interactions and molecular dynamics in ultracold quantum gases as a means to overcome uncontrolled AI shifts associated with the gravity gradient and few-particle collisions. We will further utilize the dual-species AI for proof-of-principle tests of systematic mitigations and phase-readout techniques for use in the next-generation of precision metrology experiments based on AIs in microgravity. Our proposed studies require the effective position invariance, long free fall times, and extremely low temperature samples uniquely available with the CAL apparatus. It is anticipated that our studies can lead to the unprecedented level of control and accuracy necessary for AIs to explore some of the most fundamental physical concepts in nature. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Our studies are designed to achieve technological advances in precision metrology that can only be realized in the microgravity environment of the Cold Atom Laboratory. We utilize the tools of ultracold atomic and molecular physics (namely Feshbach resonances) for exquisite control of the differential center-of-mass distributions of the dual-species quantum gases and on methods to use the fundamentals of few-body interactions to maintain coherence in atomic ensembles for enhanced precision sensor capabilities. Subsequent proof-of-principle studies with the dual-species atom interferometer on CAL will further advance the state of the art for precision interferometry with ultracold matter waves. The impact of such research to the field of metrology can be seen through its potential to increase precision for atom-interferometry and also the possibility of engineering highly efficient system-specific devices based on the fundamental nature of few-body interactions. The microgravity environment of the CAL facility will strongly favor such explorations and allow for the possibility of uncovering novel effects and quantum phases of matter, a major goal in ultracold quantum gases and other disciplines of fundamental physics. These studies can benefit life on Earth by providing both fundamental understanding of nature in previously inaccessible environments and energy regimes, and by enhancing the tools available for scientific exploration at the highest precision. |
| **Task Progress:** | The flight experiments, supported by theoretical investigations and measurements using the ground test bed facilities at JPL, will concentrate on the physics of pairwise interactions and low-energy s-wave Feshbach molecules in ultracold quantum gases as a means to overcome uncontrolled AI shifts associated with the differential center of mass of two atomic species influenced by gravity gradients and rotations. We will further utilize the dual-species AI, already in build at JPL and expected to be integrated into CAL by early 2020, for proof-of-principle demonstrations of unprecedented atom-photon coherence times, phase-readout techniques, and characterizations of the rotational noise on the ISS for use in the next-generation of precision metrology experiments based on AIs in microgravity. Our proposed experiments require the effective position invariance, long free fall times, and extremely low temperature samples uniquely available with the CAL apparatus. It is anticipated that these studies can lead to the unprecedented level of control and accuracy necessary for future space missions, based on precision AIs, to test some of the most fundamental questions of modern physics.<br>In this fifth year of performance, we have focused on the studies of molecular association using magnetic field ramps; this analysis has been finalized. This represents an alternative method to the one we have previously proposed (within this program) using RF fields and a manuscript covering such methodology is in preparation. We ultimately want to determine which of the methods for association and dissociation is more efficient in the microgravity environment. Our main finding is that a simple ramp scheme might not be sufficient to create a substantial number of molecules in the CAL's microgravity environment. We, however, further propose a scheme to fix this deficiency and verify that a much better molecular association efficiency can be expected.<br><br>Although the theoretical studies are nearly complete, experimental efforts have been hampered during this year due to required efforts by key group members to assist the CAL project in the Engineering Testbed (EMTB), Ground Test Bed (GTB), and Flight System throughout the first year of CAL's operation onboard the ISS. Significant milestones of the team that will help enable our flight project include: Leading the efforts that demonstrated CAL's minimum mission success criteria during the three-month commissioning phase after flight, demonstrated laser-cooling of 39-K using flight-hardware in the flight system and loading of potassium onto the atom-chip using flight-like hardware in the EMTB, and development of a new AI capable science module that will enable dual-species atom interferometry in the flight system in 2020.<br><br>During the sixth year, our ongoing work on this project will concentrate on working with the CAL Ground Test Bed (GTB), Engineering Testbed (EMTB), and CAL Flight Systems towards a) validating the flight hardware and b) proving all of the functionality of CAL for proof of principle and characterization studies to support our flight science projects. Due to the technical innovations required in our project and the sensitivity to numerous experimental/environmental parameters, access to the GTB has and will be enabling to mature our studies and to optimize our utilization of CAL. |
| **Bibliography Type:** | Description: (Last Updated: 05/30/2024) |
| **Abstracts for Journals and Proceedings** | D'Incao J, Williams J. "Formation of heteronuclear Feshabch molecules in microgravity." Quantum Gases forum. Presented at 49th Annual Meeting of the APS Division of Atomic, Molecular and Optical Physics (DAMOP) APS Meeting, Ft Lauderdale, FL, May 28-June 1, 2018.<br>Bulletin of the American Physical Society. 2018 Jun;63(5): Abstract T01.00102.<br>http://meetings.aps.org/Meeting/DAMOP18/Session/T01.102 , Jun-2018 |

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| **Abstracts for Journals and Proceedings** | Williams J, D'Incao J. "Opportunities for Maturing Precision Metrology with Ultracold Gas Studies Aboard the ISS." Precision Measurements and Atom Interferometers forum. Presented at 49th Annual Meeting of the APS Division of Atomic, Molecular and Optical Physics (DAMOP) APS Meeting, Ft Lauderdale, FL, May 28-June 1, 2018. Bulletin of the American Physical Society. 2018 Jun;63(5): Abstract Q06.00002. http://meetings.aps.org/Meeting/DAMOP18/Session/Q06.2 , Jun-2018 |
| **Abstracts for Journals and Proceedings** | Williams J, D'Incao J. "Maturing Space-Based Precision Metrology with Quantum Gas Studies Aboard the ISS." Atom Interferometers. Presented at NASA Fundamental Physics Workshop, La Jolla, CA, April 9-11, 2018. NASA Fundamental Physics Workshop, La Jolla, CA, April 9-11, 2018. , Apr-2018 |
| **Abstracts for Journals and Proceedings** | Williams J. "Maturing Space-Based Precision Metrology with Quantum Gas Studies Aboard the ISS." H0.6. Presented at Committee on Space Research (COSPAR) 2018 42nd Scientific Assembly, Pasadena, CA, July 14-22, 2018. Committee on Space Research (COSPAR) 2018 42nd Scientific Assembly, Pasadena, CA, July 14-22, 2018. , Jul-2018 |